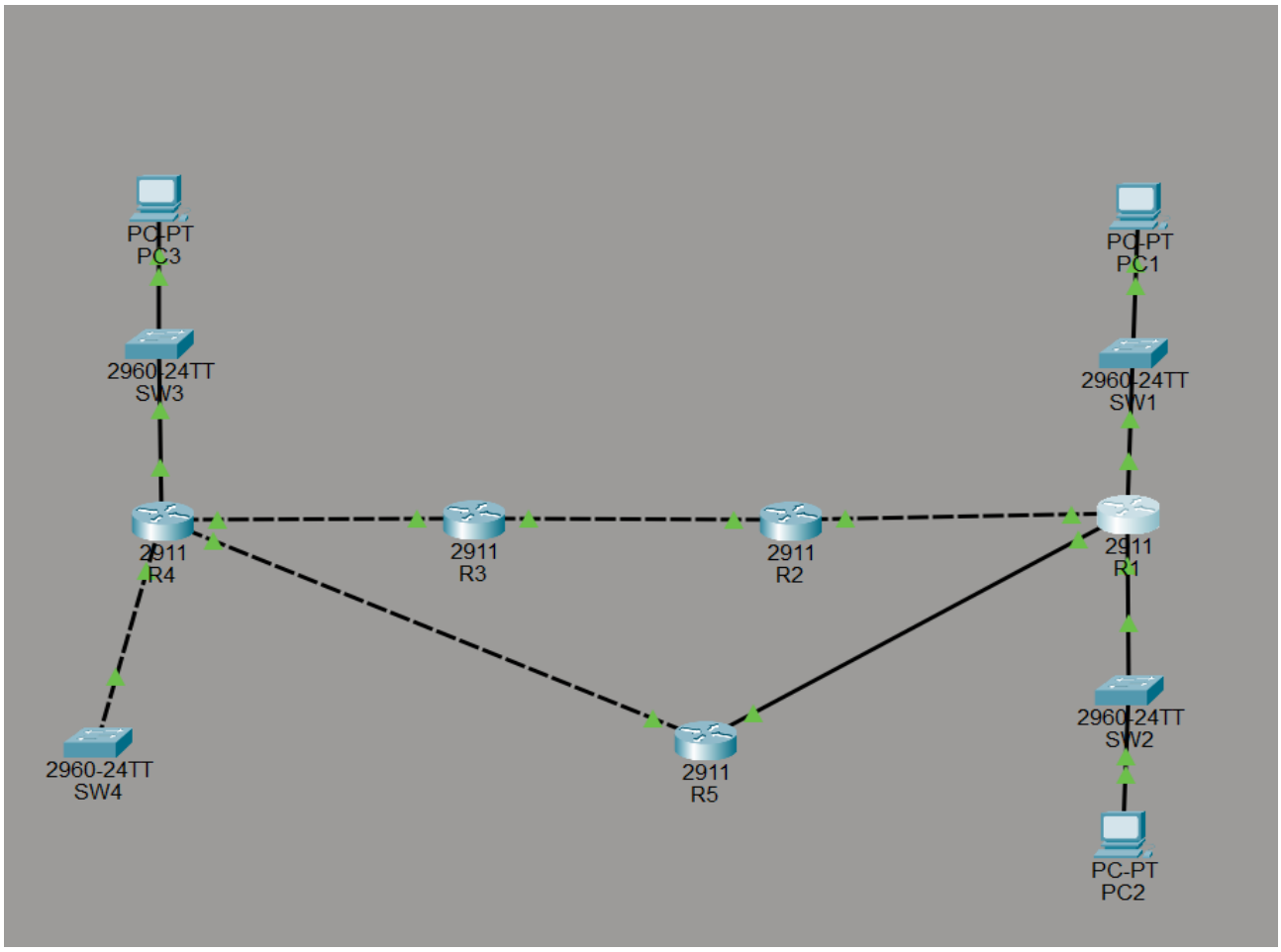
Week 8: Dynamic Routing Protocols



Configure RIPv1 on Every Router

R1

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)# network 10.0.0.0
R1(config-router)# no auto-summary
R1(config-router)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

R2

```
R2>EN
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router rip
R2(config-router)# network 10.0.0.0
R2(config-router)# no auto-summary
R2(config-router)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

R3

```
R3>EN
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router rip
R3(config-router)# network 10.0.0.0
R3(config-router)# no auto-summary
R3(config-router)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

R4

```
R4>EN
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#router rip
R4(config-router)# network 10.0.0.0
R4(config-router)# no auto-summary
R4(config-router)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console
```

R5

```
Router>EN
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Observe RIP Updates on R1

On **R1**:

```
(10.0.1.1)
RIP: build update entries
      network 10.0.0.0 metric 1
      network 10.0.2.0 metric 1
      network 10.0.3.0 metric 1
      network 10.1.0.0 metric 4
      network 10.1.1.0 metric 3
      network 10.1.2.0 metric 3
      network 10.1.3.0 metric 2
RIP: sending  v1 update to 255.255.255.255 via Vlan2 (10.0.2.1)
RIP: build update entries
      network 10.0.0.0 metric 1
      network 10.0.1.0 metric 1
      network 10.0.3.0 metric 1
      network 10.1.0.0 metric 4
      network 10.1.1.0 metric 3
      network 10.1.2.0 metric 3
      network 10.1.3.0 metric 2

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/3/1 (2), with Switch FastEthernet0/2 (1).
```

Observation

RIPv1 uses: Broadcast = 255.255.255.255

Upgrade to RIPv2

On **every router**:

```
Router>EN
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)# version 2
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Observe Traffic Type Again

R1 is still debugging.

You should now see: RIP: sending v2 update to 224.0.0.9

Observation

RIPv2 uses: Multicast = 224.0.0.9

Turn debugging off.

Confirm R1 Learns Every Network

On R1: Check Routing Table After OSPF Converges

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set


Router#show ip rip database
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set


Router#
```

Copy Paste

Disable R2 FastEthernet 0/0

Using your topology, this corresponds to the interface connecting R2 to R1.

```
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface g0/0
R2(config-if)# shutdown

R2(config-if)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to
administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to down

R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.1.0.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.0.3.0/24 [110/4] via 10.1.0.1, 00:01:46, GigabitEthernet0/1
C       10.1.0.0/24 is directly connected, GigabitEthernet0/1
L       10.1.0.2/32 is directly connected, GigabitEthernet0/1
S       10.1.1.0/24 [1/0] via 10.1.0.1
S       10.1.2.0/24 [1/0] via 10.1.0.1
S       10.1.3.0/24 [1/0] via 10.1.0.1
S*   0.0.0.0/0 [1/0] via 10.1.0.1

R2#
```

Prediction

R1 loses the path through:

R1 → R2 → R3 → R4

and uses:

R1 → R5 → R4

View OSPF Database

```
Router#show ip ospf database
%OSPF: Router process 1 is not running, please configure a router-id
```

Reflection Answers

11. What kind of traffic is used for RIPv1 updates, and how does this change with RIPv2?

RIPv1 sends updates using **broadcast traffic (255.255.255.255)**. RIPv2 sends updates using the **multicast address 224.0.0.9**, reducing unnecessary processing by devices that are not running RIP.

12. Why are there two routes to the 10.1.1.0/24 network before OSPF is configured?

There are two valid RIP paths: R1 → R2 → R3 → R4 and R1 → R5 → R4

Both routes reach the same destination network, so RIP can learn multiple paths.

13. Once OSPF converges, are RIP routes still present in R1's routing table? Why or why not and why is there now only one route to 10.1.1.0/24?

RIP routes are no longer installed in the routing table because OSPF has a lower Administrative Distance (110) than RIP (120). OSPF is therefore preferred. Only the best OSPF route is installed for 10.1.1.0/24.

14. After disabling FastEthernet 0/0 on R2, what changed on R1's routing table besides the next-hop address?

The OSPF metric (cost) and path calculation changed. R1 now reaches remote networks through R5 and R4 instead of through R2 and R3.

15. What is the key difference between the RIP database and the OSPF database, and why does that difference exist?

The RIP database contains only destination networks and hop-count information. The OSPF database contains a complete link-state view of the topology, including routers, links, and network relationships. This difference exists because RIP is a distance-vector protocol while OSPF is a link-state protocol.